Lecture 15: Nonlinear Resistive Circuits

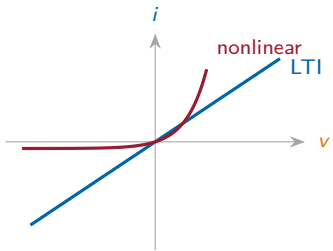
EE201 – Circuit Theory I

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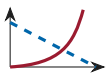
current voltage element characteristic - - - load line

Review: an element is a curve



- A resistor: a curve in the v–i plane
- LTI: a line through the origin
- Nonlinear: any other curve, e.g. $i = g(v)$
- So far: LTI circuits only

Roadmap



Load line

1

operating point

none or many



Substitution

2

Thévenin first

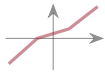


Ideal diode

3

on and off

series, parallel



Piecewise-linear circuits

4

i–*v* of a circuit

circuit from *i*–*v*

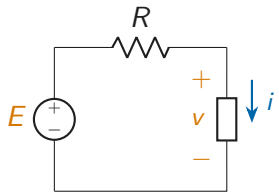


Time-varying

5

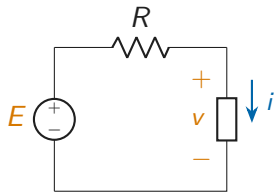
superposition

Motivation: nonlinear elements



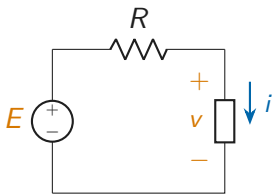
- Diodes, lamps, transistors: nonlinear
- Node and mesh analysis: linear equations

Motivation: nonlinear elements



- Diodes, lamps, transistors: nonlinear
- Node and mesh analysis: linear equations
- Now: one nonlinear equation, or a few linear pieces

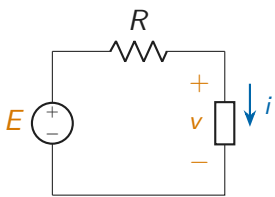
One nonlinear resistor



KVL: $E = R\,i + v$

Element: $i = g(v)$

One nonlinear resistor



Element: $i = g(v)$

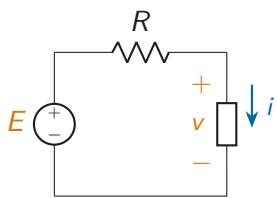
KVL: $E = R\,i + v$

Algebraic solution

Solve $E = R\,g(v) + v$ for v_o

Then $i_o = g(v_o)$

One nonlinear resistor



Element: $i = g(v)$

KVL: $E = R\,i + v$

Algebraic solution

Solve $E = R\,g(v) + v$ for v_o

Then $i_o = g(v_o)$

Often g is only a measured curve

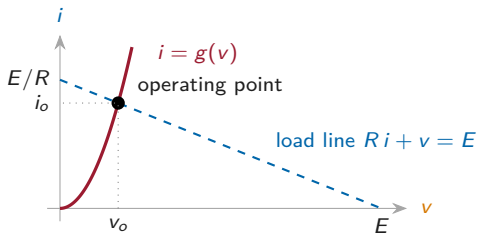
The load line

Two constraints on (v, i):

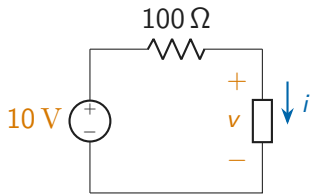
circuit: $R\,i + v = E$

element: $i = g(v)$

The solution is on both



Example 15.1: the operating point



$$i = \begin{cases} \frac{3}{100}\, v^2, & v \geq 0 \\ 0, & v < 0 \end{cases}$$

$$(v_o, i_o) = ?$$

Example 15.1: two cases

KVL: $100\,i + v - 10 = 0$

$v < 0$**:** $i = 0 \;\Rightarrow\; v = 10$

contradiction: no solution

Example 15.1: two cases

KVL: $100\,i + v - 10 = 0$

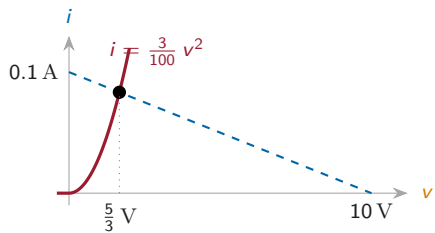
$v < 0$: $i = 0 \;\Rightarrow\; v = 10$

contradiction: no solution

$v \geq 0$: $3\,v^2 + v - 10 = 0$

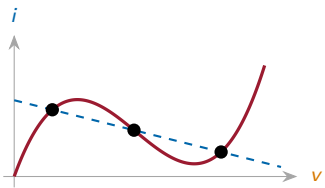
$(3v - 5)(v + 2) = 0 \;\Rightarrow\; v = \frac{5}{3}\,\mathrm{V}$

$$(v_o, i_o) = \left(\tfrac{5}{3}\,\mathrm{V},\ \tfrac{1}{12}\,\mathrm{A}\right)$$



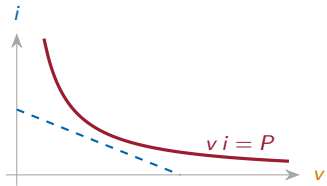
None, one or many operating points

Many



A bent curve: three intersections

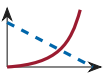
None



No intersection

A nonlinear circuit need not have exactly one solution

Roadmap



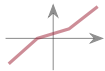
Load line



Substitution



Ideal diode



Piecewise-linear circuits



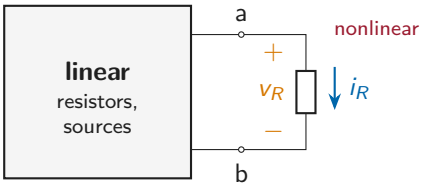
Time-varying

✓ 2 3 4 5

operating point ✓ | **Thévenin first** | on and off | i–v of a circuit | superposition

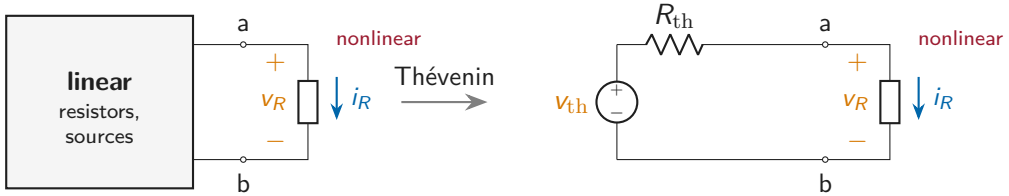
none or many ✓ | series, parallel | circuit from i–v

A nonlinear resistor in a larger circuit



One nonlinear resistor; the rest of the circuit is linear

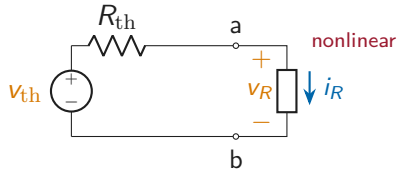
A nonlinear resistor in a larger circuit



One nonlinear resistor; the rest of the circuit is linear

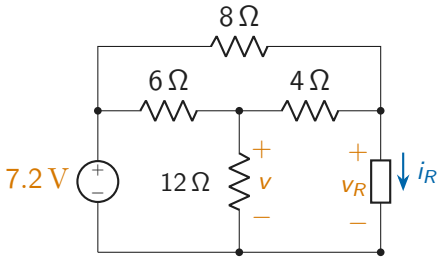
Seen from a–b, the rest is a source and a resistor: the case we just solved

Three steps



- **Step 1:** the Thévenin equivalent seen from a–b
- **Step 2:** the load line gives the operating point (v_R, i_R)
- **Step 3:** substitution theorem: replace the nonlinear resistor by a source v_R (or i_R)
- No other voltage or current changes
- Any other voltage or current then follows from a linear circuit

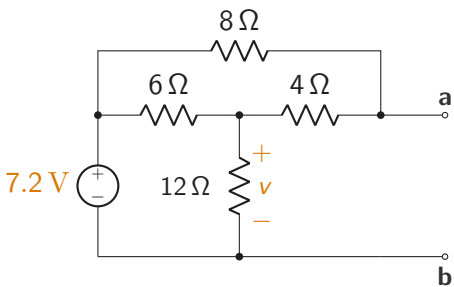
Example 15.2: find v



$$i_R = \begin{cases} \frac{1}{4} {v_R}^2, & v_R \geq 0 \\ 0, & v_R < 0 \end{cases}$$

$v = ?$

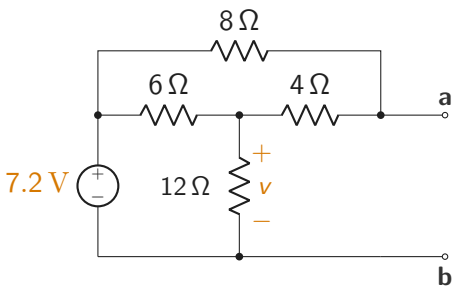
Example 15.2: step 1, Thévenin



Kill the source:

$R_{\text{th}} = 8 \parallel (4+6 \parallel 12) = 8 \parallel 8 = 4\,\Omega$

Example 15.2: step 1, Thévenin

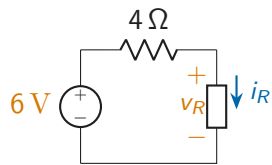


Kill the source:

$R_{\text{th}} = 8 \parallel (4+6 \parallel 12) = 8 \parallel 8 = 4\,\Omega$

Open circuit: two node equations

$v_{\text{th}} = v_{\text{oc}} = 6\,\text{V}$



Example 15.2: step 2, the operating point

KVL: $4\, i_R + v_R - 6 = 0$

Plug $i_R = \frac{1}{4}\, {v_R}^2$:

${v_R}^2 + v_R - 6 = 0$

$(v_R + 3)(v_R - 2) = 0$

Example 15.2: step 2, the operating point

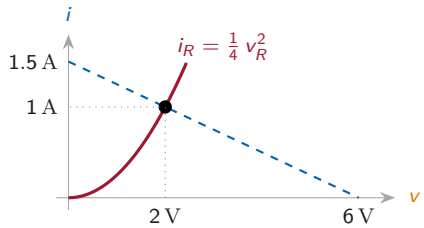
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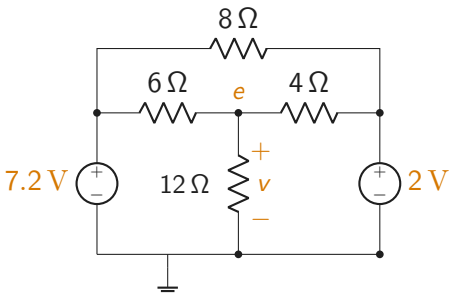
${v_R}^2 + v_R - 6 = 0$

$(v_R + 3)(v_R - 2) = 0$

$v_R \geq 0$: $v_R = 2\,\mathrm{V}$, $i_R = 1\,\mathrm{A}$



Example 15.2: step 3, substitution

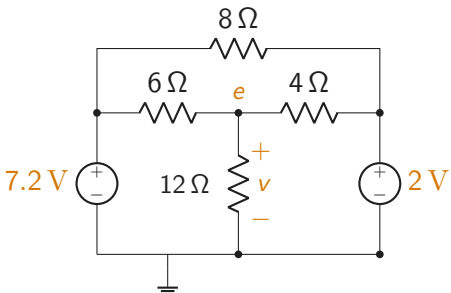


Replace it by a 2 V source

Node *e*:

$$\frac{e-7.2}{6}+\frac{e}{12}+\frac{e-2}{4}=0$$

Example 15.2: step 3, substitution



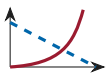
Replace it by a $2\,\mathrm{V}$ source

Node *e*:

$$\frac{e-7.2}{6}+\frac{e}{12}+\frac{e-2}{4}=0$$

$$v = e = 3.4\,\mathrm{V}$$

Roadmap



Load line ✓

operating point ✓

none or many ✓



Substitution ✓

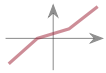
Thévenin first ✓



Ideal diode 3

on and off

series, parallel



Piecewise-linear circuits 4

i–*v* of a circuit

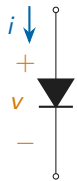
circuit from *i*–*v*

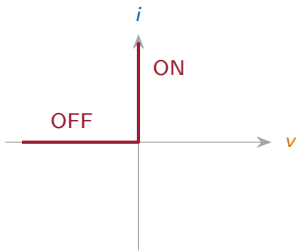


Time-varying 5

superposition

The ideal diode





ON: $v = 0, \quad i \geq 0$

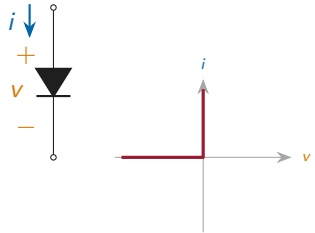
a short circuit

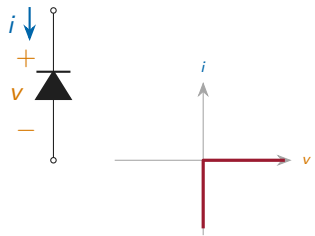
OFF: $i = 0, \quad v \leq 0$

an open circuit

Two straight pieces:
piecewise linear

Turning an element around





Swap the terminals: $v \to -v$, $i \to -i$

The curve is reflected through the origin

Series and parallel

Series: the same i

add the voltages

$v = v_a + v_b$

add the curves horizontally

Parallel: the same v

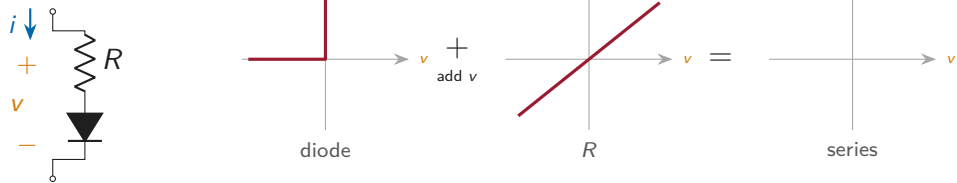
add the currents

$i = i_a + i_b$

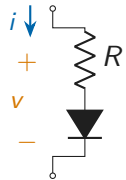
add the curves vertically

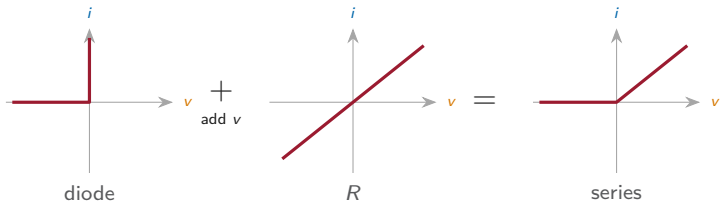
Diodes, LTI resistors and constant sources: the result is piecewise-linear

Case I: a resistor and a diode in series



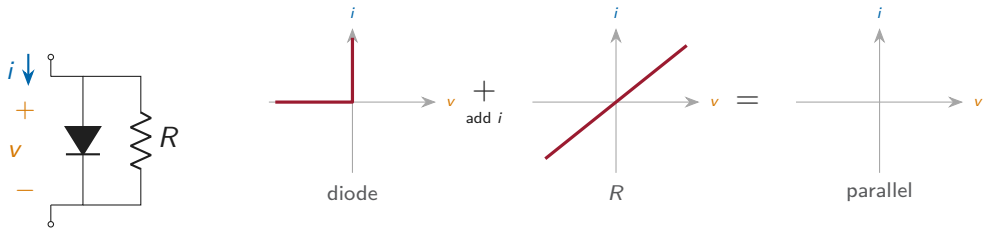
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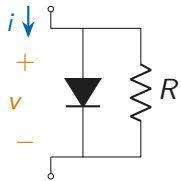


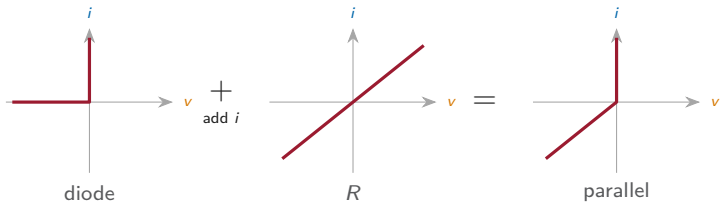
$$i > 0: \quad v = R\,i; \qquad i = 0: \quad v \leq 0$$

Case II: a resistor and a diode in parallel



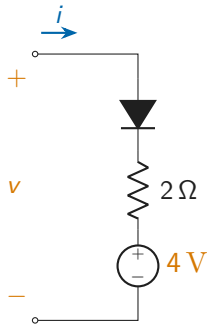
Case II: a resistor and a diode in parallel





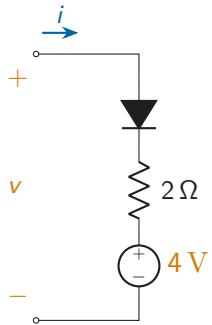
$v < 0$: $i = v/R$; $v = 0$: any $i \geq 0$

Example 15.3: a diode, a resistor and a source



i as a function of v $= ?$

Example 15.3: a diode, a resistor and a source



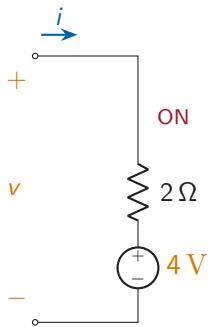
i as a function of v $= ?$

Method: assume a state for the diode

Solve the linear circuit

Keep the result where the assumption holds

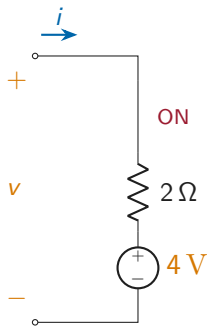
Example 15.3: assume the diode ON



ON: a short; condition $i \geq 0$

KVL: $v = 2\,i + 4$

Example 15.3: assume the diode ON

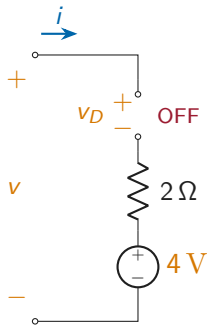


ON: a short; condition $i \geq 0$

KVL: $v = 2\,i + 4$

$$i = \frac{v - 4}{2} \geq 0 \;\Leftrightarrow\; v \geq 4\,\mathrm{V}$$

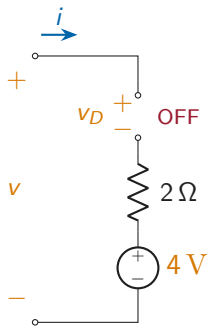
Example 15.3: assume the diode OFF



OFF: an open; condition $v_D < 0$

$i = 0$: no voltage on $2\,\Omega$

Example 15.3: assume the diode OFF



OFF: an open; condition $v_D < 0$

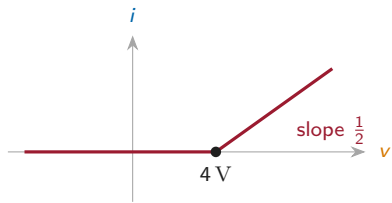
$i = 0$: no voltage on $2\,\Omega$

KVL: $v_D = v - 4$

$v_D < 0 \;\Leftrightarrow\; v < 4\,\mathrm{V}$

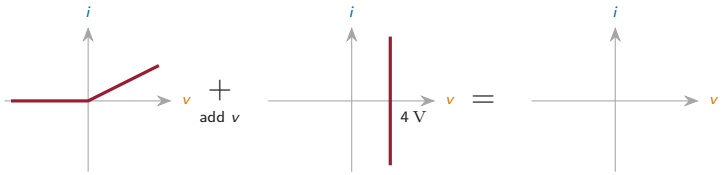
Example 15.3: the result

$$i = \begin{cases} \dfrac{v-4}{2}, & v \geq 4\,\mathrm{V} \\ 0, & v < 4\,\mathrm{V} \end{cases}$$

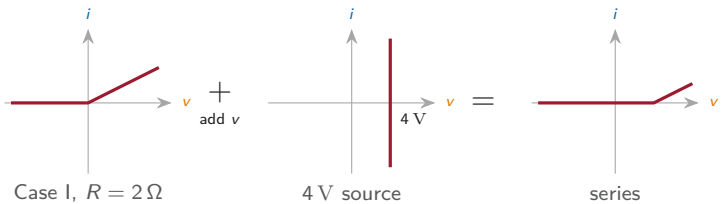


Breakpoint: $i_D = 0$ and $v_D = 0$

Example 15.3: graphically

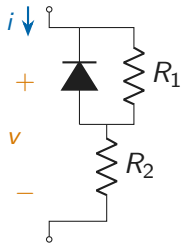


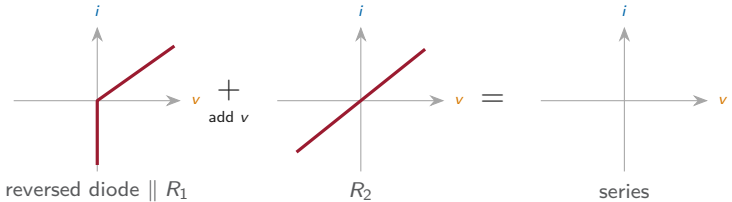
Example 15.3: graphically



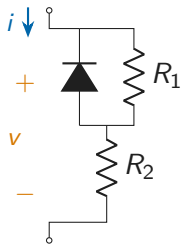
A source in series shifts the curve right by $4\,\mathrm{V}$ ✓

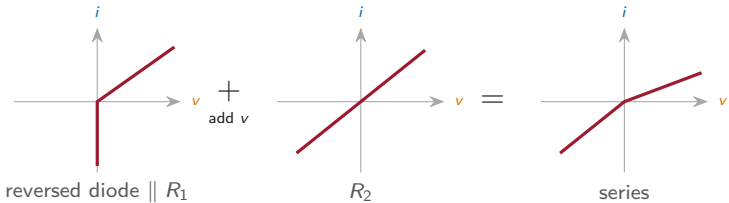
Case III: combinations





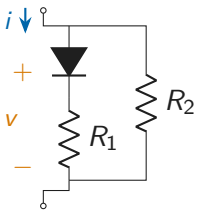
Case III: combinations

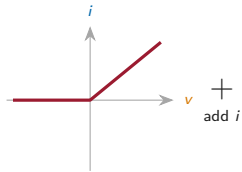




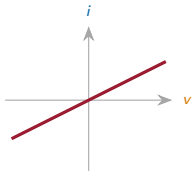
$i > 0$: diode OFF, slope $\dfrac{1}{R_1 + R_2}$; $\quad i < 0$: diode ON, slope $\dfrac{1}{R_2}$

Case IV: combinations

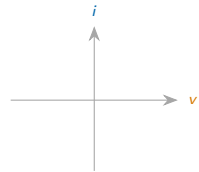




diode, R_1 in series

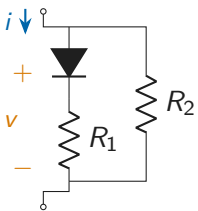


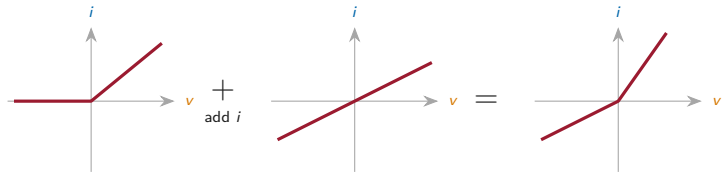
R_2



parallel

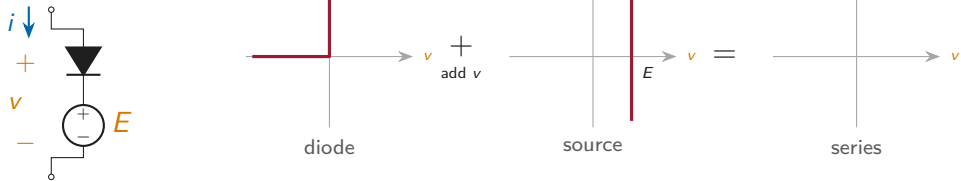
Case IV: combinations



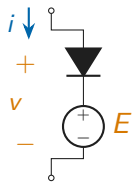


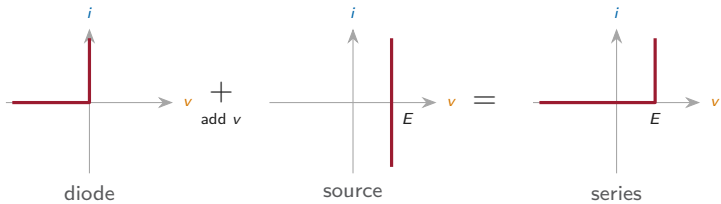
$$v > 0\text{: slope } \frac{1}{R_1} + \frac{1}{R_2}; \qquad v < 0\text{: slope } \frac{1}{R_2}$$

Case V: a diode and a voltage source in series



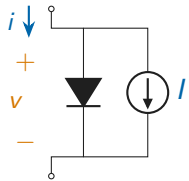
Case V: a diode and a voltage source in series

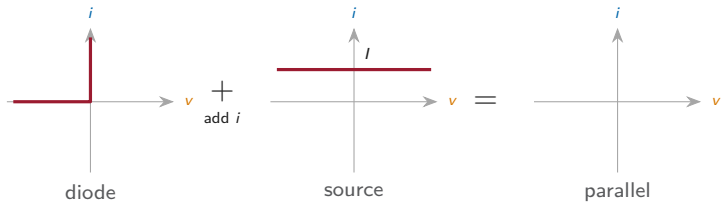




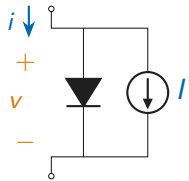
The diode curve shifted right by E

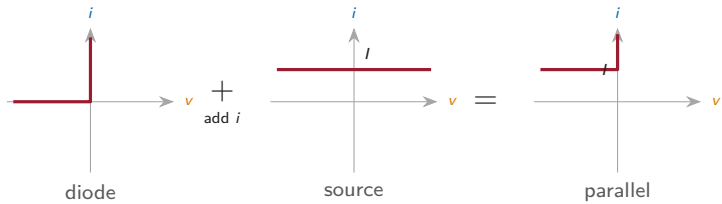
Case VI: a diode and a current source in parallel





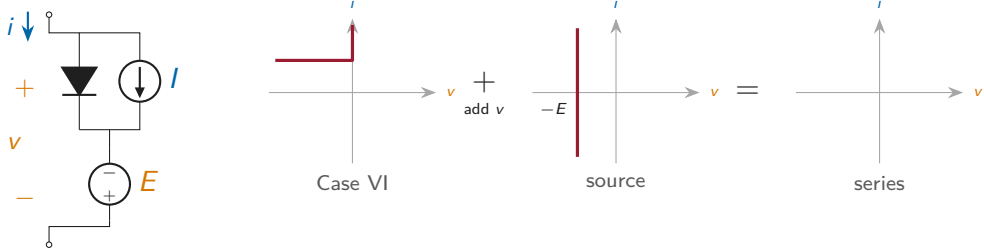
Case VI: a diode and a current source in parallel



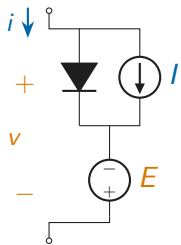


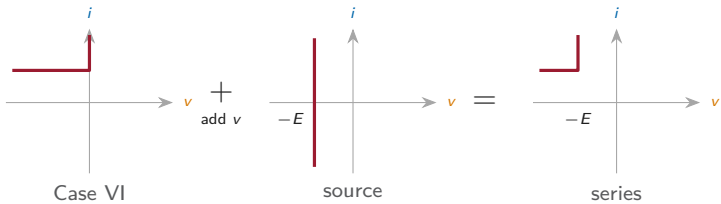
The diode curve shifted up by I

Case VII: sources and a diode



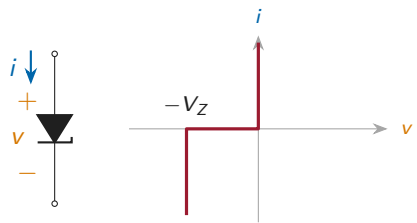
Case VII: sources and a diode





The source has $+$ at the bottom: shifted left by E

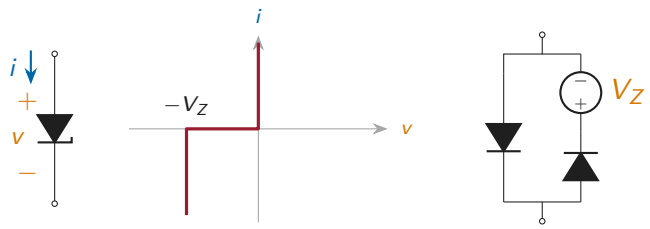
The ideal Zener diode



Forward: like a diode

Reverse: blocks until $v = -V_Z$

The ideal Zener diode

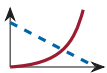


Forward: like a diode

Reverse: blocks until $v = -V_Z$

Two diodes and a battery V_Z

Roadmap



Load line ✓

operating point ✓

none or many ✓



Substitution ✓

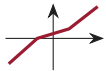
Thévenin first ✓



Ideal diode ✓

on and off ✓

series, parallel ✓



Piecewise-linear circuits 4

i–*v* of a circuit

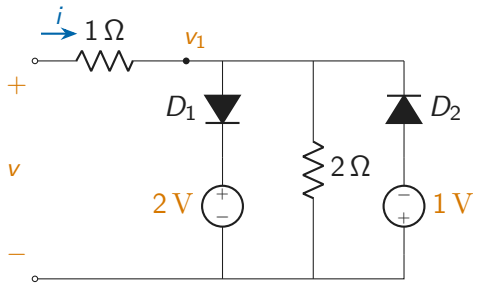
circuit from *i*–*v*



Time-varying 5

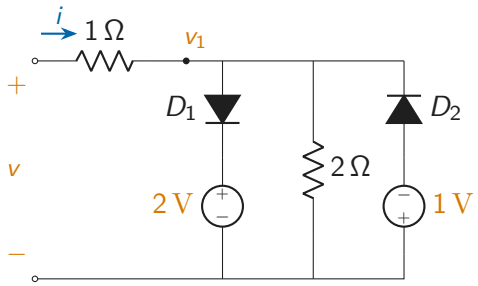
superposition

Example 15.4: two diodes



i as a function of v $= ?$

Example 15.4: two diodes



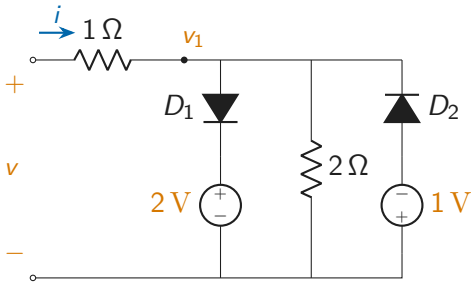
i as a function of v $= ?$

Two diodes: four combinations

One linear circuit for each

Keep each result where it is valid

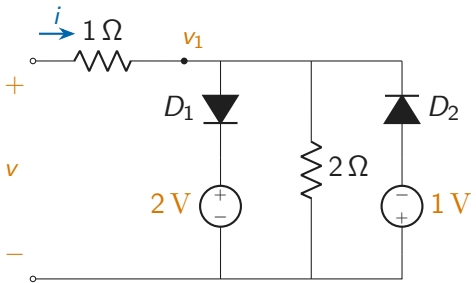
Example 15.4: both diodes OFF



1 Ω and 2 Ω in series:

$v_1 = 2\,i, \quad v = 3\,i$

Example 15.4: both diodes OFF



1 Ω and 2 Ω in series:

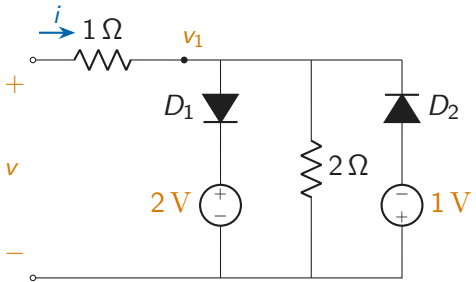
$v_1 = 2\,i, \quad v = 3\,i$

D_1 OFF: $v_{D_1} = v_1 - 2 < 0$

D_2 OFF: $v_{D_2} = -1 - v_1 < 0$

$-1 < v_1 < 2 \;\Leftrightarrow\; -1.5 < v < 3$

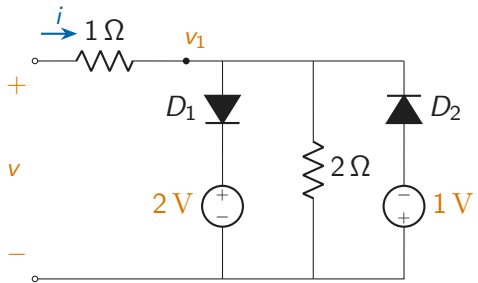
Example 15.4: D_1 ON, D_2 OFF



D_1 a short: $v_1 = 2$

$v = i + 2$

Example 15.4: D_1 ON, D_2 OFF



D_1 a short: $v_1 = 2$

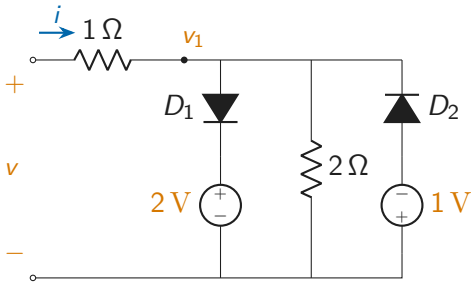
$v = i + 2$

$2\,\Omega$: $1\,\mathrm{A}$; D_1: $i_{D_1} = i - 1$

$i_{D_1} \geq 0 \;\Leftrightarrow\; v \geq 3$

D_2: $v_{D_2} = -1 - 2 < 0$ ✓

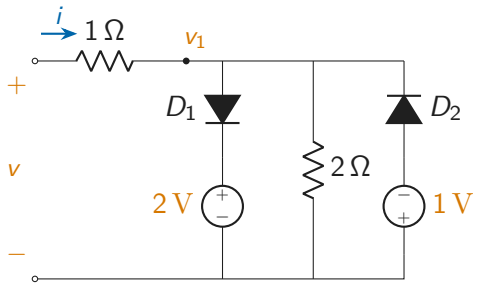
Example 15.4: D_2 ON, D_1 OFF



D_2 a short: $v_1 = -1$

$v = i - 1$

Example 15.4: D_2 ON, D_1 OFF



D_2 a short: $v_1 = -1$

$v = i - 1$

$2\,\Omega$: $-0.5\,\mathrm{A}$; D_2: $i_{D_2} = -0.5 - i$

$i_{D_2} \geq 0 \;\Leftrightarrow\; v \leq -1.5$

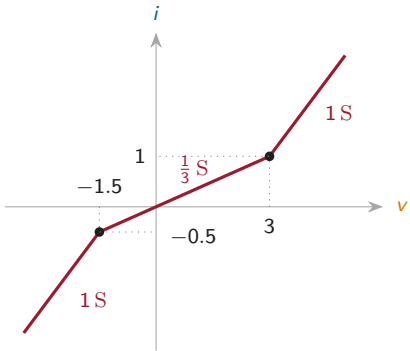
D_1: $v_{D_1} = -1 - 2 < 0$ ✓

Example 15.4: the result

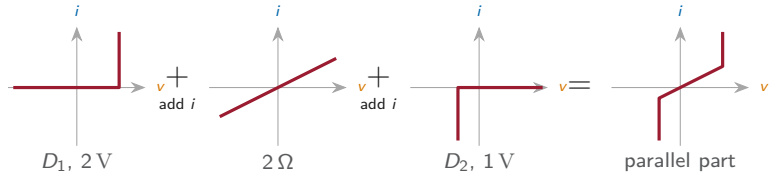
Both ON: $v_1 = 2$ and -1 (impossible)

$$i = \begin{cases} v - 2, & v \geq 3 \\ v/3, & -1.5 \leq v \leq 3 \\ v + 1, & v \leq -1.5 \end{cases}$$

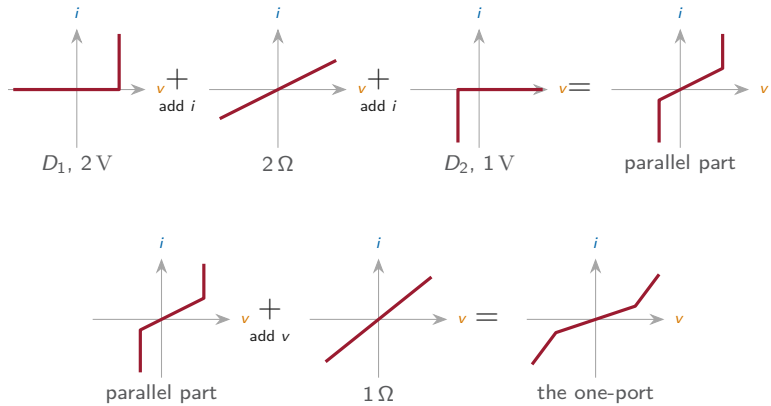
Two diodes: two breakpoints



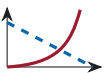
Example 15.4: graphically



Example 15.4: graphically



Roadmap



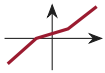
Load line



Substitution



Ideal diode



Piecewise-linear circuits



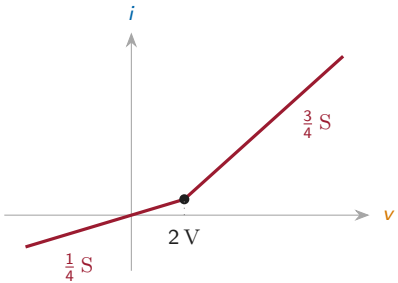
Time-varying

✓ ✓ ✓ 4 5

operating point ✓ | Thévenin first ✓ | on and off ✓ | *i*–*v* of a circuit ✓ | superposition

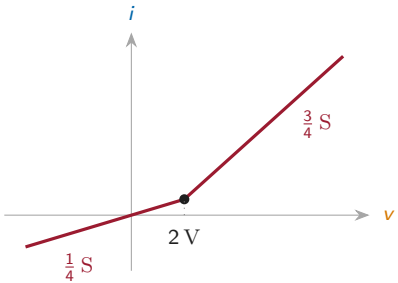
none or many ✓ | series, parallel ✓ | **circuit from *i*–*v***

Example 15.5: a circuit from a characteristic



A circuit with this i–v characteristic $=$?

Example 15.5: a circuit from a characteristic

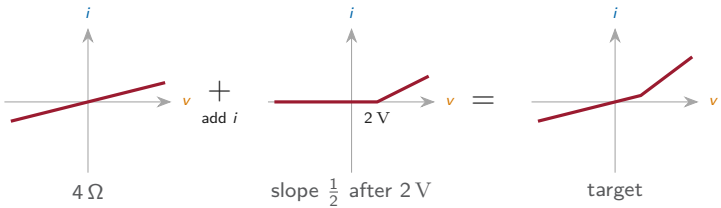


A circuit with this i–v characteristic = ?

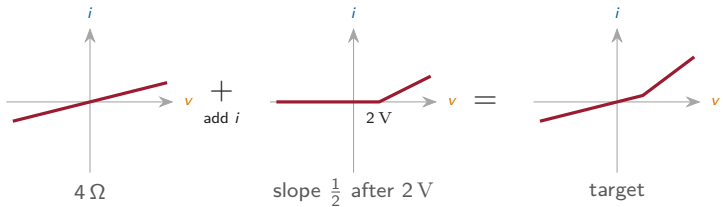
One breakpoint: one diode

Below $2\,\mathrm{V}$: a line through the origin

Example 15.5: split the characteristic

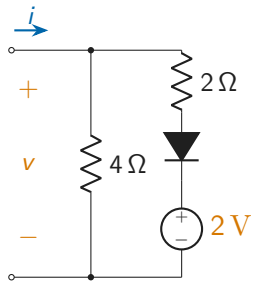


Example 15.5: split the characteristic



- Slope $\frac{1}{4}\,\mathrm{S}$ everywhere: a $4\,\Omega$ resistor
- Extra slope $\frac{3}{4} - \frac{1}{4} = \frac{1}{2}\,\mathrm{S}$ after $2\,\mathrm{V}$: Case I with $2\,\Omega$, shifted by $2\,\mathrm{V}$
- Currents add: the two in parallel

Example 15.5: the circuit

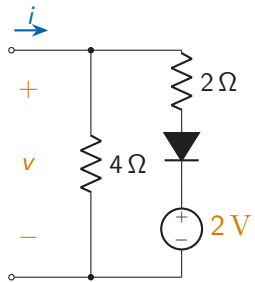


Check at $v = 4\,\mathrm{V}$:

$4\,\Omega$: $1\,\mathrm{A}$; branch: $\dfrac{4-2}{2} = 1\,\mathrm{A}$

$i = 2\,\mathrm{A}$ ✓

Example 15.5: the circuit



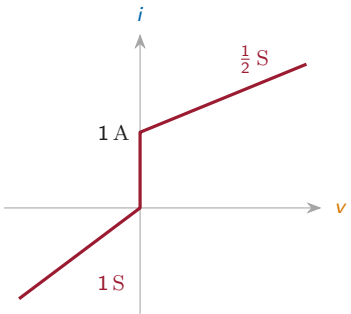
Check at $v = 4\,\mathrm{V}$:

$4\,\Omega$: $1\,\mathrm{A}$; branch: $\dfrac{4-2}{2} = 1\,\mathrm{A}$

$i = 2\,\mathrm{A}$ ✓

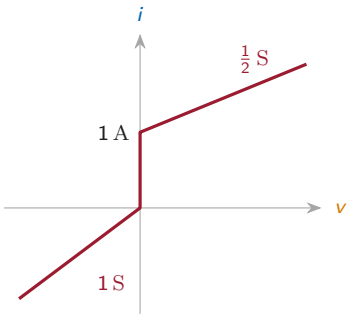
The order inside the branch does not matter: the circuit is not unique

Example 15.6: a circuit from a characteristic



A circuit with this i–v characteristic = ?

Example 15.6: a circuit from a characteristic

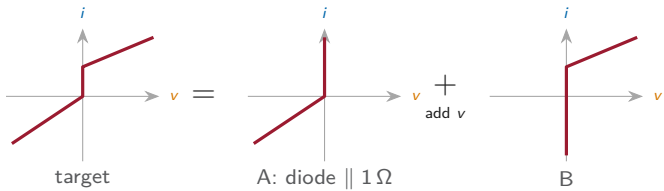


A circuit with this i–v characteristic = ?

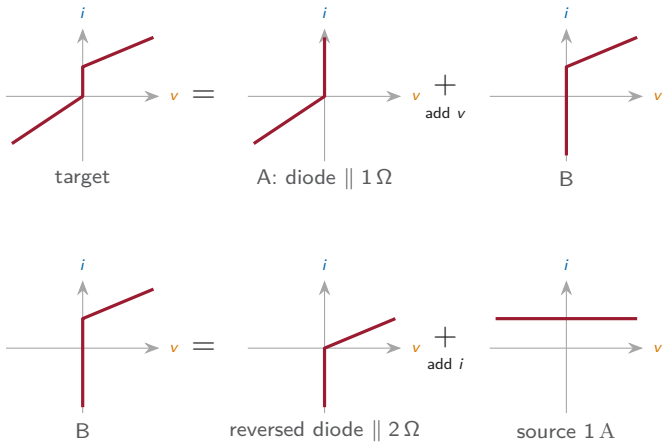
Two breakpoints: at least two diodes

Split into pieces we know

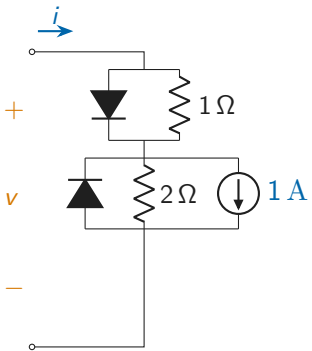
Example 15.6: split the characteristic



Example 15.6: split the characteristic



Example 15.6: the circuit



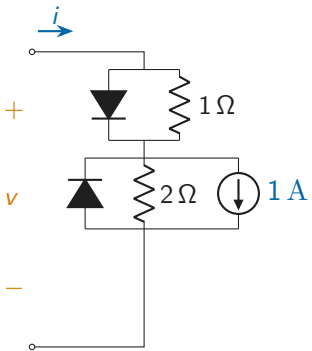
Top: A, the diode $\| \, 1\,\Omega$

Bottom: B, reversed diode $\| \, 2\,\Omega \| \, 1\,\mathrm{A}$

A and B in series

Check at $v = 2\,\mathrm{V}$: $i = 1 + \frac{2}{2} = 2\,\mathrm{A}$ ✓

Example 15.6: the circuit



Top: A, the diode $\parallel 1\,\Omega$

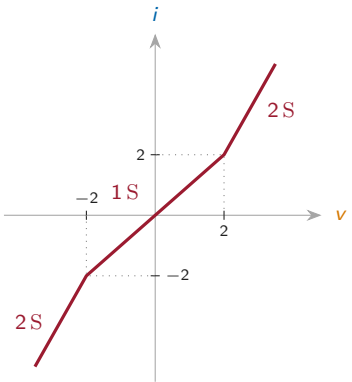
Bottom: B, reversed diode $\parallel 2\,\Omega \parallel 1\,\mathrm{A}$

A and B in series

Check at $v = 2\,\mathrm{V}$: $i = 1 + \frac{2}{2} = 2\,\mathrm{A}$ ✓

Another circuit may have the same characteristic: the circuit is not unique

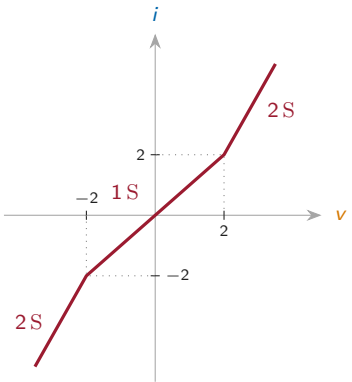
Example 15.7: a numerical characteristic



Slopes $2\,\mathrm{S}$, $1\,\mathrm{S}$, $2\,\mathrm{S}$

A circuit $=$?

Example 15.7: a numerical characteristic



Slopes $2\,\mathrm{S}$, $1\,\mathrm{S}$, $2\,\mathrm{S}$

A circuit $=$?

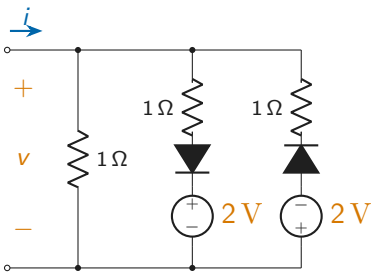
Split (parallel, add currents):

a $1\,\Omega$ resistor

$+$ a $1\,\mathrm{S}$ branch for $v > 2\,\mathrm{V}$

$+$ a $1\,\mathrm{S}$ branch for $v < -2\,\mathrm{V}$

Example 15.7: the circuit

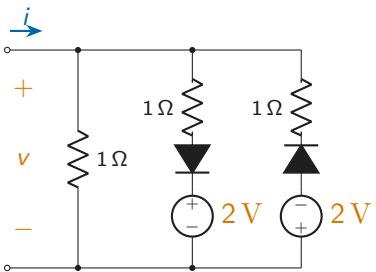


Branch 2: $v - 2$ for $v > 2\,\mathrm{V}$

Branch 3: turned around

$v + 2$ for $v < -2\,\mathrm{V}$

Example 15.7: the circuit



Branch 2: $v - 2$ for $v > 2\,\mathrm{V}$

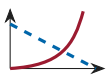
Branch 3: turned around

$v + 2$ for $v < -2\,\mathrm{V}$

Check at $v = 3\,\mathrm{V}$:

$3 + 1 + 0 = 4\,\mathrm{A}$ ✓

Roadmap



Load line

operating point ✓

none or many ✓



Substitution

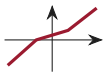
Thévenin first ✓



Ideal diode

on and off ✓

series, parallel ✓



Piecewise-linear circuits

i–v of a circuit ✓

circuit from i–v ✓

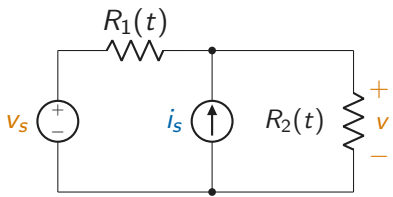


Time-varying

5

superposition

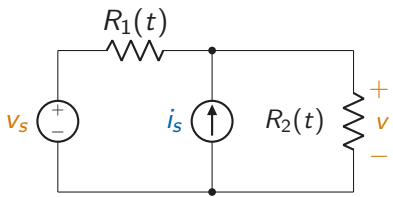
Example 15.8: a linear time-varying circuit



$R_1(t)$, $R_2(t)$: linear, time-varying

KCL: $\dfrac{v - v_s}{R_1} - i_s + \dfrac{v}{R_2} = 0$

Example 15.8: a linear time-varying circuit

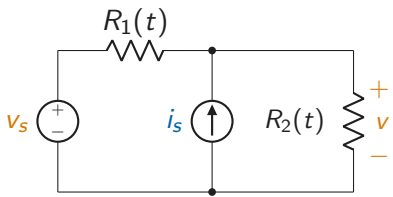


$R_1(t)$, $R_2(t)$: linear, time-varying

KCL: $\dfrac{v - v_s}{R_1} - i_s + \dfrac{v}{R_2} = 0$

$$v = \underbrace{\frac{R_2}{R_1 + R_2}}_{k_1(t)} v_s + \underbrace{\frac{R_1 R_2}{R_1 + R_2}}_{k_2(t)} i_s$$

Example 15.8: a linear time-varying circuit



$R_1(t)$, $R_2(t)$: linear, time-varying

KCL: $\dfrac{v - v_s}{R_1} - i_s + \dfrac{v}{R_2} = 0$

$$v = \underbrace{\frac{R_2}{R_1 + R_2}}_{k_1(t)} v_s + \underbrace{\frac{R_1 R_2}{R_1 + R_2}}_{k_2(t)} i_s$$

$v = \mathbf{k}(t)\,\mathbf{u}$: superposition still holds

Summary

In the circuit	What to do
one nonlinear resistor	load line $R\,i + v = E$ meets $i = g(v)$
a linear rest	Thévenin, operating point, substitution
ideal diodes	assume ON / OFF, solve, check
series, parallel	add voltages, add currents
a given characteristic	split it into known pieces
linear time-varying	$v = \mathbf{k}(t)\,\mathbf{u}$: superposition holds